

ELEMENT B

PROBLEM EVALUATION & ELABORATION

Design Attempt	Successful / Unsuccessful	Strength	Weaknesses
Box Inlet Drop Spillway	Was shown to be successful on a large scale, but for small scales, the design is unable to be moved and is really for a large body of water.	Has a stable structure to withstand water erosion or other environmental factors, a lower chance of the design being filled with debris or other objects that may clog it, and is simple to build and construct.	Expensive to build, not suited for something temporary. Can be eroded much faster when not built correctly, in a poor location where the ground is already being eroded.
Condensate Overflow Safety Switch	Unsuccessful; the design needs a power source in order to be used, and the switch can get worn out in a humid climate.	Allows it to be shut off when the water level exceeds a certain point and poses a safety hazard to the surroundings.	Requires a consistent electrical supply, allowing it to damage the circuit. Can get worn out due to humid environments that Hawaii is located in. If not installed correctly, the AC won't send a shut-off signal in time, allowing the water to drip inside the AC. If the user doesn't clean regularly, debris can build up, clogging the line and causing the triggering system to occur often.
Rubber tubing	Successful: the design allows it to be connected to anything and be put in any direction the user needs.	Can be flexed into any direction to a certain area. Allows the AC to transport its refrigerant somewhere else. Can withstand AC's vibrations, pressure changes, and temperature swings.	If not installed properly, it can be hard for the AC to set the correct temperature that was set. Can be broken easily and doesn't last a long time if made from something cheap. If not installed properly, it can lead to air or water leaking.
Condensate Drain Trap	Successful, the device only takes up a small amount of space yet takes the water that the AC drips without missing it.	Occupies a small area of space, and removable caps on both ends allowed it to be cleaned or inspected easily. Can shut down the AC unit to prevent more condensation water from coming and leaking.	Sudden cut off of electricity supply can lead to compressor strain, electrical surge, and interruption of cooling cycle. Without cleaning it regularly, it can lead to moisture and mold, worsening the AC and device.
Drain Pan Assembly	Successful; allows for collecting a large area where the AC is dripping, collects a large amount depending on how big the area is.	Can be made out of anything that can be folded and waterproof. Can be made without a manual on how to make it. Does not leak from the corners due to being folded.	Standing water: insects can lay eggs there; disease-carrying species can spread dengue and malaria. Potential waterlogging, causing the water to leak everywhere, leading to the floor being slippery.

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